Appendix H - Per-CPU Symbol Index

Entry points, ABI conventions, and reserved memory regions for each CPU. The detailed register descriptions and the per-CPU chapter give the full story; this appendix is the cheat sheet.

H.1 IE64

Symbol	Value / role
Reset vector	`$000000` (first instruction at start of RAM).
Trap vector base	`$000400` (`8`-byte entries, indexed by trap number).
Supervisor stack	grows down from `$0A0000`.
User stack (`R31`)	grows down from BASIC's per-program stack region.
Call ABI	Arguments `R8`-`R15`; result `R8`; caller-saved `R1`-`R7`; callee-saved `R16`-`R30`; `R0 = 0`; `R31 = SP`.
FPU regs	`F0`-`F15`; FP32 values, with double operations using register pairs. `F0`-`F7` are argument / result registers by convention.
BASIC text	`$023000`-`$04FFFF` (`BASIC_PROG_START`-`BASIC_PROG_LIMIT - 1`).
Simple vars	`$050000`-`$057FFF`.
String vars	`$058000`-`$05FFFF`.
Arrays	`$060000`-`$08BFFF`.
String temps	`$08C000`-`$08FFFF`.
GOSUB / FOR stack	`$090000`-`$096FFF`.

H.2 IE32

Symbol	Value / role
Reset vector	`$000000`.
Stack base	`$09F000` (`STACK_START`); grows down.
Timing	`WAIT n` for short microsecond delays; use device status or interrupts for frame and audio timing.
Call ABI	Arguments A,X,Y,Z; result A; B-W caller-saved; stack via PUSH / POP.

H.3 6502

Symbol	Value / role
Reset vector	`$FFFC` (low) / `$FFFD` (high).
IRQ / BRK vector	`$FFFE` / `$FFFF`.

Symbol	Value / role
NMI vector	`$FFFA` / `$FFFB`.
Stack page	`$0100-$01FF`, indexed by `S`.
Zero page	`$0000-$00FF`.
Bank registers	`$F700-$F705`, `$F7F0`.
MMIO aperture	`$F000-$FFF9`, mirrors `$F0000-$F0FF9`.
VGA C64-style	`$D700-$D70D`.
ULA paged port	`$D800-$D817`.
PSG / SID	`$D400-$D40F`, `$D500-$D55F`.
POKEY	`$D200-$D20A`.
TED audio	`$D600-$D605`.

H.4 Z80

Symbol	Value / role
Reset vector	`$0000`.
NMI vector	`$0066`.
RST n	`n * 8` (`$00`, `$08`, ..., `$38`).
IM 2 vector base	`(I << 8) | (data byte from device)`.
Bank registers	port-mapped through bus translation at `$F700-$F705`.
MMIO aperture	`$F000-$FFF9`.
PSG / AY ports	`$F0` select, `$F1` data.
TED audio ports	`$F2`, `$F3`.
POKEY ports	`$D0`, `$D1`.
SID ports	`$E0`, `$E1`.
SN76489 ports	`$E4` data, `$E5` status.
VGA ports	`$A0-$AD`.

H.5 M68K (MC68020-Class)

Symbol	Value / role
Reset vector	`$0000.0000` (initial SSP) / `$0000.0004` (initial PC).
Vector table	`$0000.0000-$0000.03FC` (256 entries, 4 bytes each).
Bus error	vector 2.
Address error	vector 3.
Illegal	vector 4.

Symbol	Value / role
Zero divide	vector 5.
CHK	vector 6.
Trapv	vector 7.
Privilege violation	vector 8.
Trace	vector 9 (trace bits are stored; this chip does not raise trace traps).
Line A	vector 10.
Line F	vector 11.
TRAP #n	vectors 32-47.
Auto-vector IRQs	vectors 25-31.
Call ABI	Arguments on stack; D0 / A0 caller-saved; D2-D7 / A2-A6 callee-saved.

H.6 x86 (8086 + 386 extensions, real-mode only)

Symbol	Value / role
Reset vector	`EIP = 0`, `CS = 0`, `DS = ES = SS = 0` (flat, not the 8086 `F000:FFF0` boot vector).
`.ie86` image start	bytes loaded at `$00000000`, execution starts with `EIP = 0`; use an entry stub at `0` if the body lives elsewhere.
IVT	`$0000`-`$03FF` (`256` entries, `4` bytes each: offset + segment).
MMIO data access	`$000F0000`-`$000FFFFF` is native MMIO. Data accesses in `$F000`-`$FFFF` mirror `$000F0000`-`$000F0FFF`. Instruction fetch at `$F000` reads flat RAM at `$0000F000`.
Stack	`SS:ESP`, segments are zero so the stack lives in flat RAM.
Call ABI	Caller pushes arguments right-to-left; `EAX` returns; `EBX`, `ESI`, `EDI`, `EBP` callee-saved; `ECX`, `EDX` caller-saved.
BIOS-style ints	reserved; no BIOS ROM is provided. The IVT is initialised to a default IRET routine.

Real-mode 20-bit physical address calculation `(seg << 4) + ofs` is part of the CPU address path. The 32-bit linear form (the result of the calculation) is what reaches the bus. Programs that use 32-bit immediate addressing reach the full flat address space directly.

H.7 Cross-CPU bus addresses (shared)

These addresses are the same in every CPU's 32-bit view of the bus. The 8-bit CPUs reach them through the bank-window mechanism described in Chapters 27 and 28.

Address	Meaning
`$F0700`	`TERM_OUT`.
`$F075C`/`$F0760`	`RTC_MONO_USEC_LO` / `RTC_MONO_USEC_HI`, monotonic elapsed microseconds.
`$F1400`	HOST appliance block.
`$F2200`	File I/O block.

Address	Meaning
`$F2300`	Media loader.
`$F2320`	RUN loader block.
`$F2340`	Coprocessor.
`$F2400`	SysInfo.
`$F8000`	Voodoo 3D.